



GEN-AI: ARTIFICIAL INTELLIGENCE AND THE FUTURE OF WORK

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AI and Human Collaboration: Shaping Future Job Markets

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Gen-AI: Artificial Intelligence and the Future of Work

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STAFF DISCUSSION NOTE

The Global Impact of AI: Mind the Gap

Eugenio Cerutti, Antonio Garcia Pascual, Yosuke Kido, Longji Li, Giovanni Melina, Marina M. Tavares, and Philippe Wingender

WP/25/76

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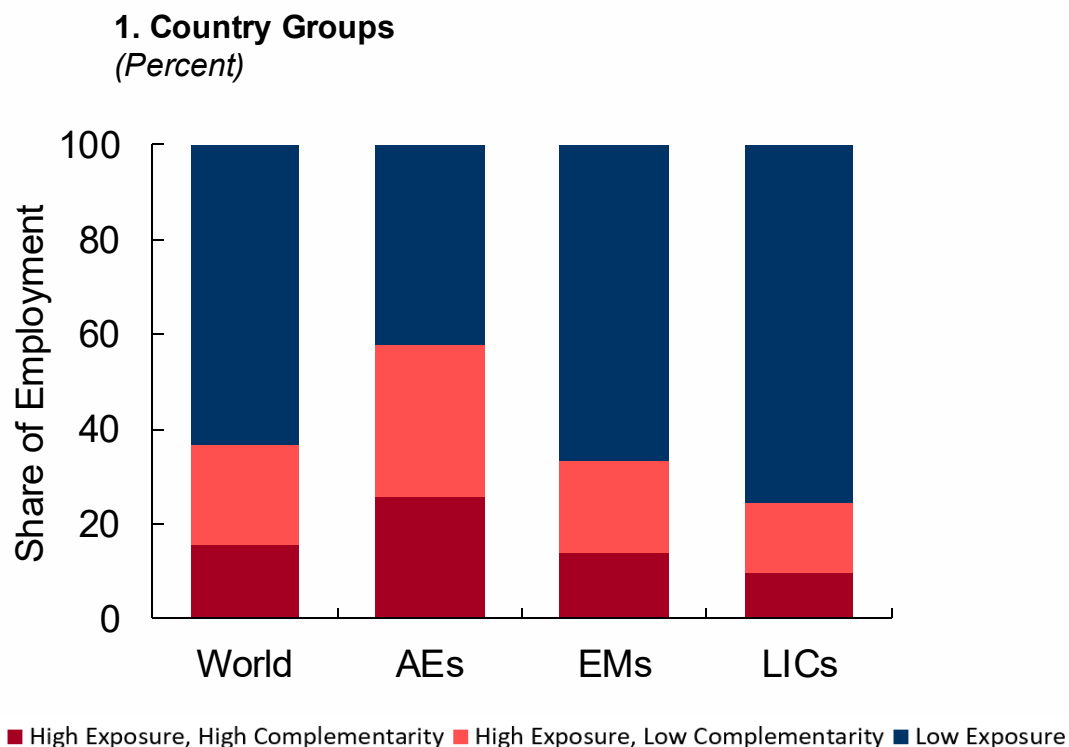
WORKING PAPER

Motivation and Focus

- Artificial Intelligence (AI) is set to profoundly change the global economy
- We examine:
 - implications of AI adoption on jobs across AEs and EMDEs
 - its potential to displace and complement human labor
 - potential effects of AI on inequality and productivity
 - countries' preparedness to adopt AI
 - unequal macroeconomic effects from AI across countries

About forty percent of workers worldwide and sixty percent in AEs is in high-exposure occupations

Employment Shares by AI Exposure and Complementarity



Sources: American Community Survey (ACS); Gran Encuesta Integrada de Hogares (GEIH); India Periodic Labour Force Survey (PLFS); International Labour Organization (ILO); Labour Market Dynamics in South Africa (LMDSA); Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC); UK Labour Force Survey (LFS); and IMF staff calculations.

Note: Country labels use International Organization for Standardization (ISO) country codes. ISCO stands for International Standard Classification of Occupations. AEs = advanced economies; EMs = emerging markets; LICs = low-income countries; World = all countries in the sample. Share of employment within each country group is calculated as the working-age-population-weighted average.

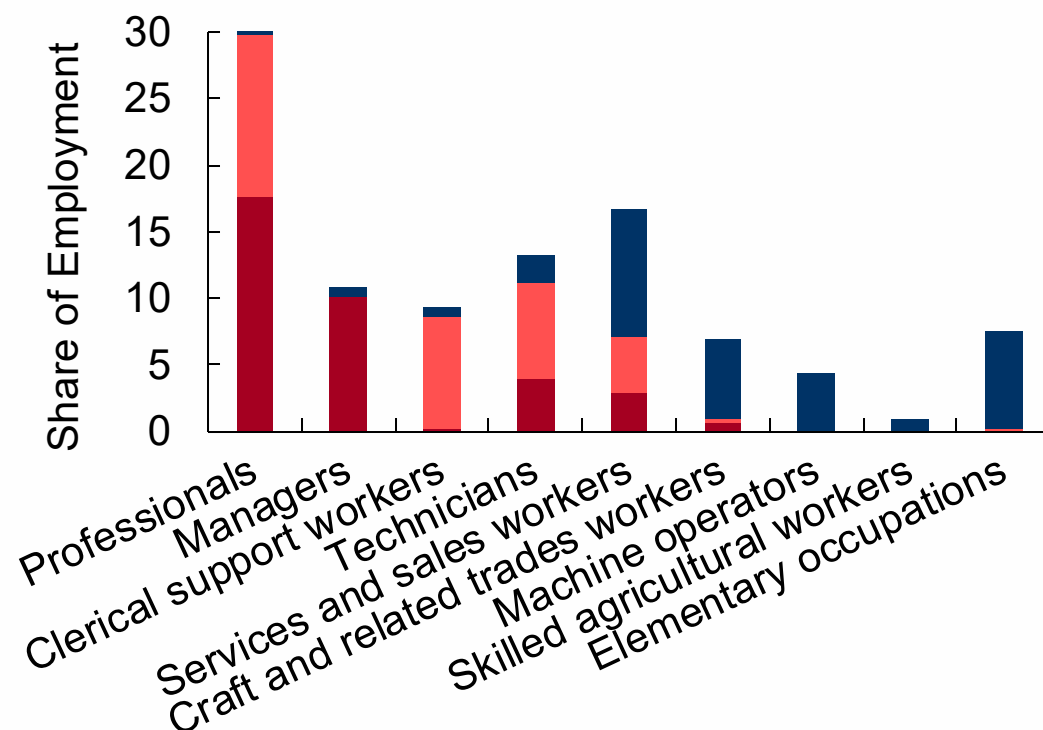
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- AI exposure and complementarity varies by income group:
 - ▶ AEs: 27% high-complementarity; 33% low complementarity jobs;
 - ▶ EMs: 16% high-complementarity; 24% low complementarity jobs;
 - ▶ LICs: 8% high-complementarity; 18% low complementarity jobs.
- AEs dominate in cognitive-intensive roles, potentially facing more immediate AI job disruption.
- However, AEs also have a stronger position to harness AI's growth potential.
- With appropriate digital infrastructure, AI could help EMDEs mitigate skill shortages.

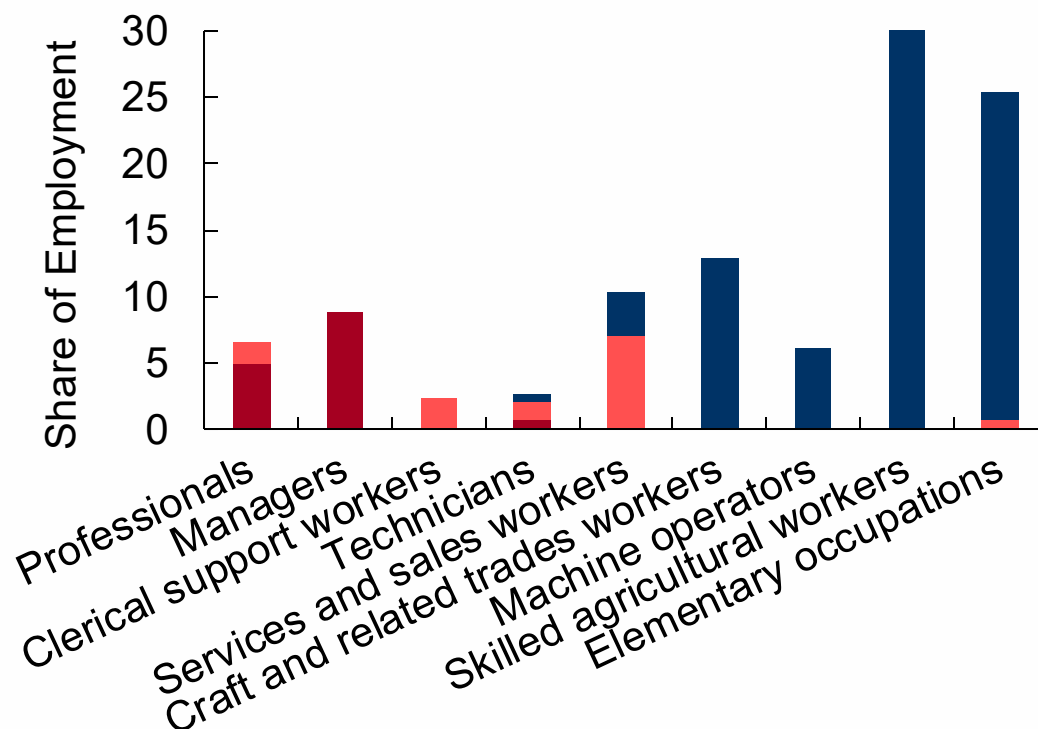
Labor force composition in terms of broad occupational groups largely explains the differences in exposure and complementarity across countries

Employment Share by Exposure and Complementarity

1. GBR
(Percent)



2. IND
(Percent)



■ High Exposure, High Complementarity ■ High Exposure, Low Complementarity ■ Low Exposure

Sources: India Periodic Labour Force Survey (PLFS); Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC); UK Labour Force Survey (LFS); and IMF staff calculations.

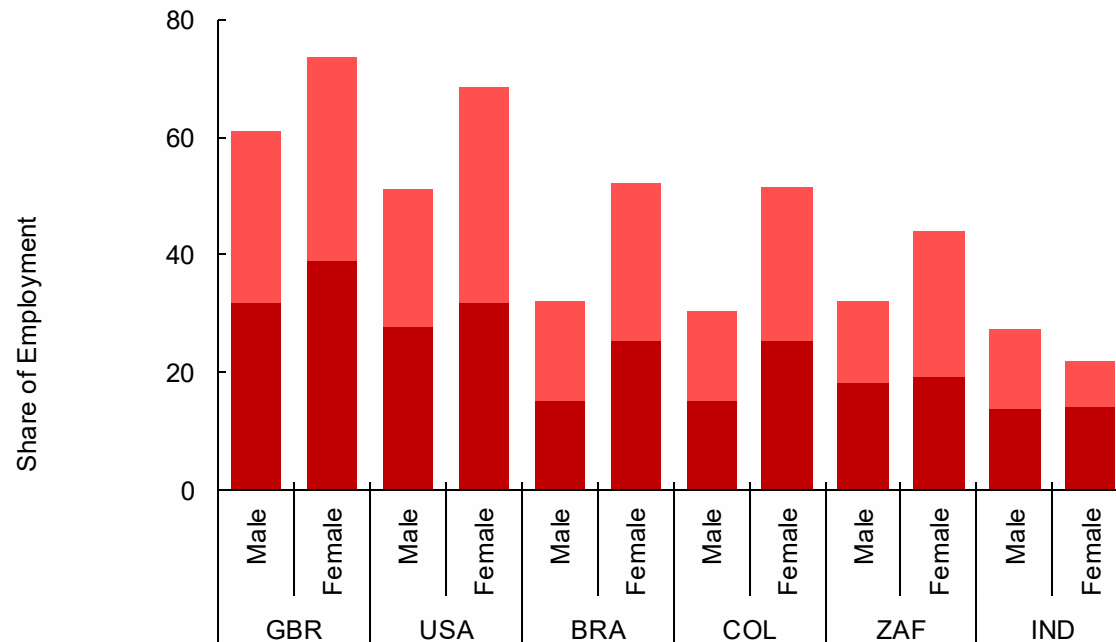
Note: The charts plot the total employment share by each of the nine 1-digit ISCO-08 occupation codes. Country names use International Organization for Standardization (ISO) country codes. ISCO stands for International Standard Classification of Occupations.

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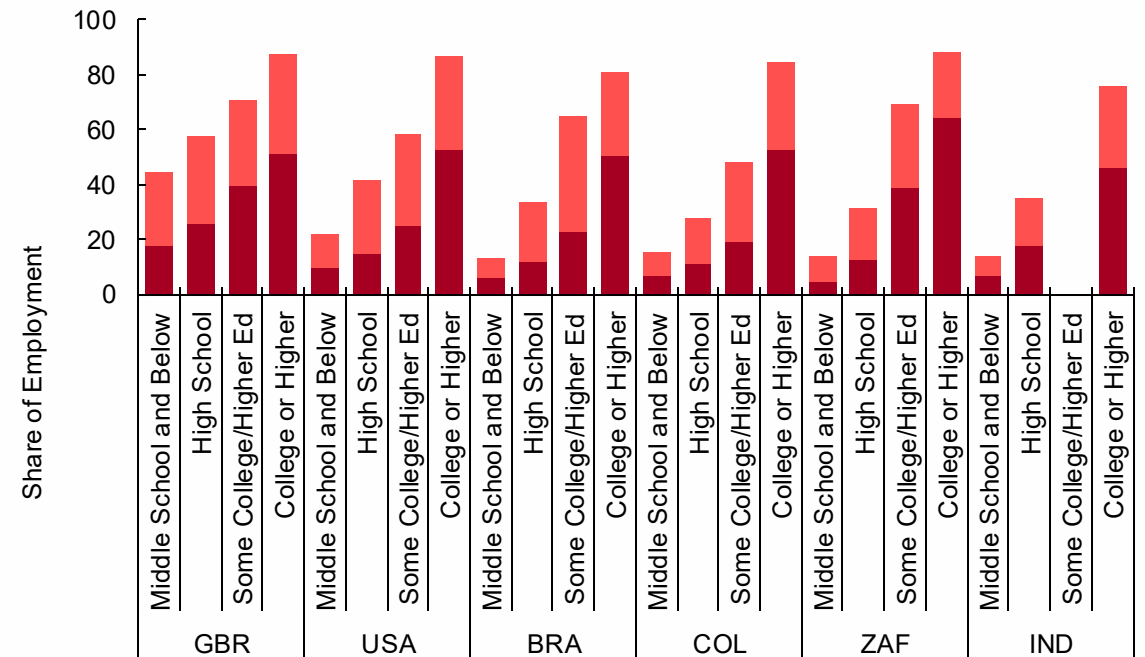
Exposure is higher for women and for more educated workers, but is mitigated by a higher potential for complementarity with AI

Share of Employment in High-Exposure Occupations by Demographic Groups

1. By Gender
(Percent)



2. By Education
(Percent)



■ High Exposure, High Complementarity

■ High Exposure, Low Complementarity

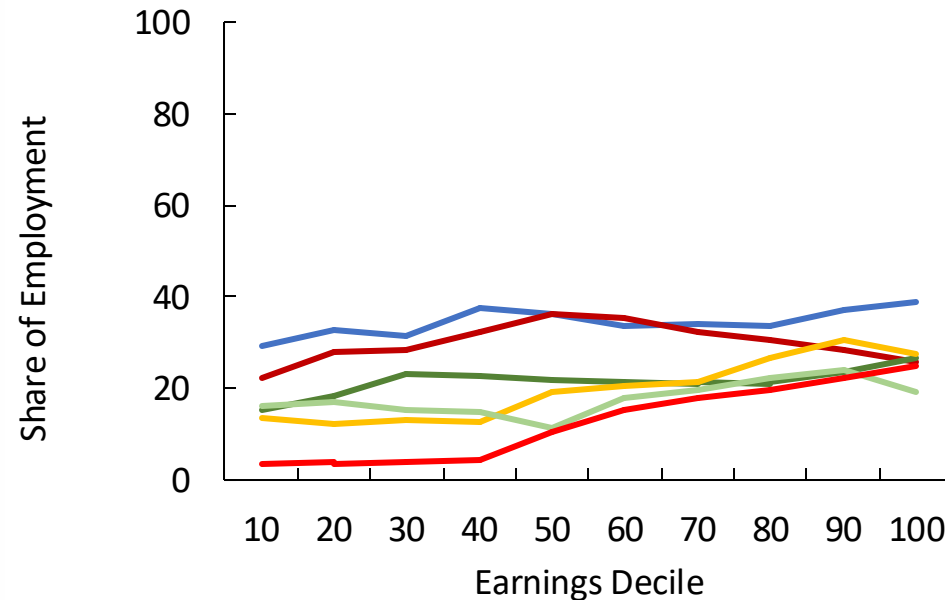
Sources: American Community Survey (ACS); Gran Encuesta Integrada de Hogares (GEIH); India Periodic Labour Force Survey (PLFS); Labour Market Dynamics in South Africa (LMDSA); Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC); UK Labour Force Survey (LFS); and IMF staff calculations.

Note: The bars in both plots represent employment shares in high-exposure occupations. In plot 1, employment shares are conditional on each gender category. In plot 2, employment shares are conditional on each of the four education categories (Middle School and Below, High School, Some College and College). In plot 3, employment shares are conditional on each of the four age intervals. Country labels use International Organization for Standardization (ISO) country codes.

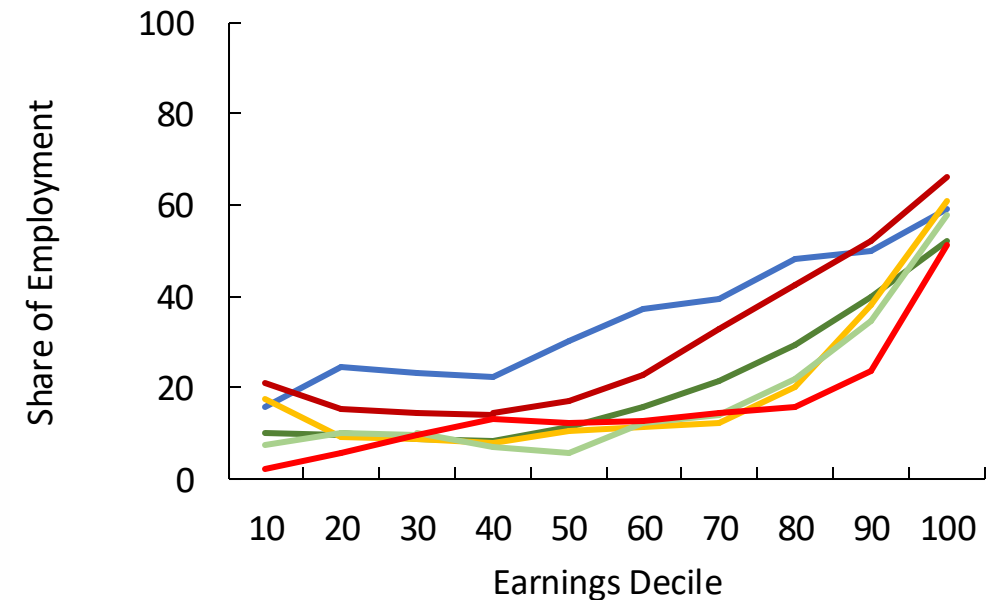
Exposure is spread along the labor income distribution but potential gains from AI are positively correlated with income

Share of Employment in High-Exposure Occupations and Potential Complementarity by Income Deciles

1. High-Exposure, Low-Complementarity
(Percent)



2. High-Exposure, High-Complementarity
(Percent)



— GBR — USA — BRA — COL — ZAF — IND

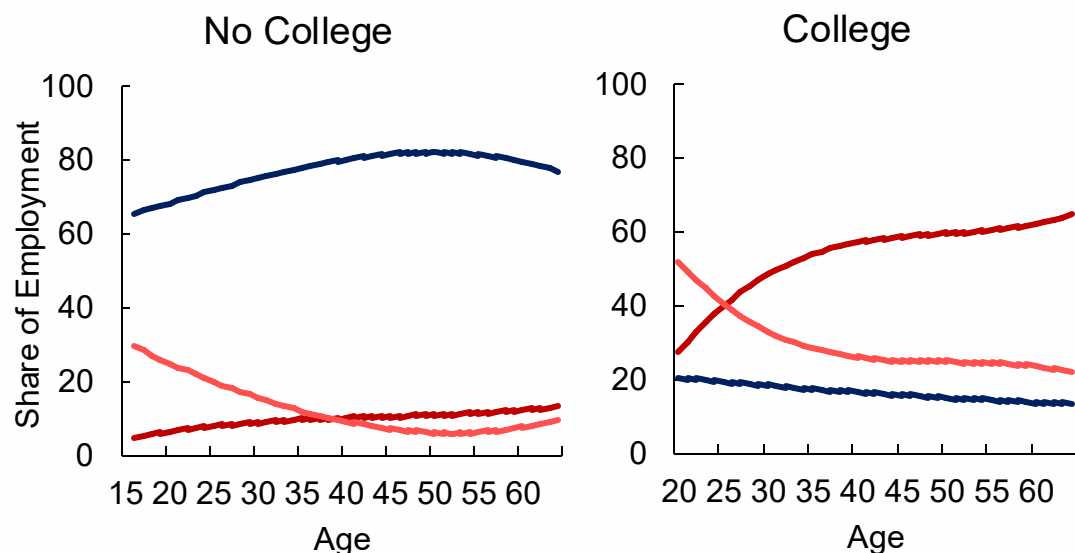
Sources: American Community Survey (ACS); Gran Encuesta Integrada de Hogares (GEIH); India Periodic Labour Force Survey (PLFS); Labour Market Dynamics in South Africa (LMDSA); Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC); Pizzinelli and others (2023); UK Labour Force Survey (LFS); and IMF staff calculations.

Note: Panel 1 shows the employment share in jobs with high exposure but low complementarity, and Panel 2 presents the employment share in jobs with high exposure and high complementarity, each categorized by income deciles. Panel 3 shows the potential AI occupational complementarity from Pizzinelli and others (2023), averaged and grouped by income deciles. Country labels use International Organization for Standardization (ISO) country codes.

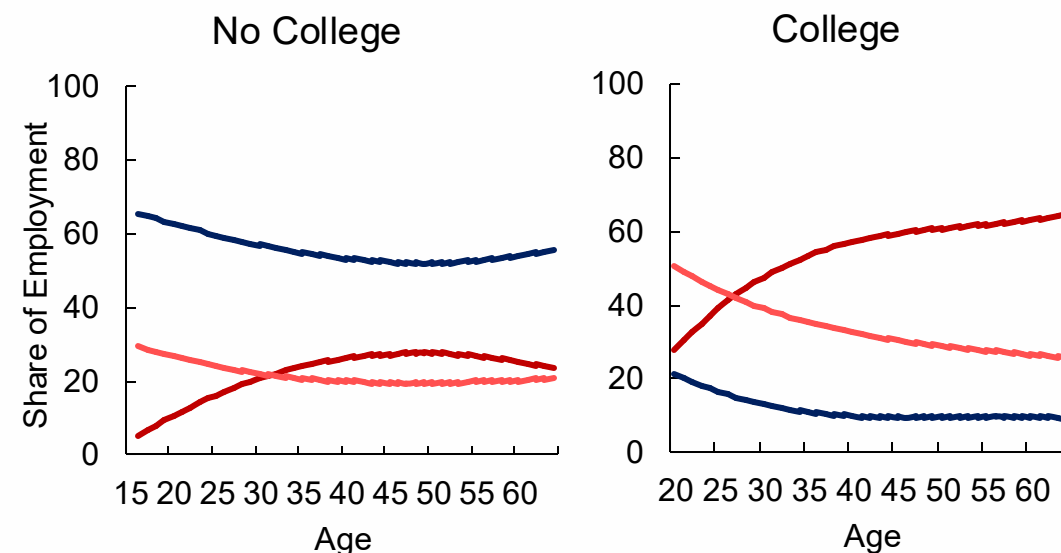
AI adoption both poses challenges and represents an opportunity for young college-educated workers' careers

Life Cycle Profiles of Employment Shares by Education Level

1. BRA (Percent)



2. GBR (Percent)



— High Exposure, High Complementarity — High Exposure, Low Complementarity — Low Exposure

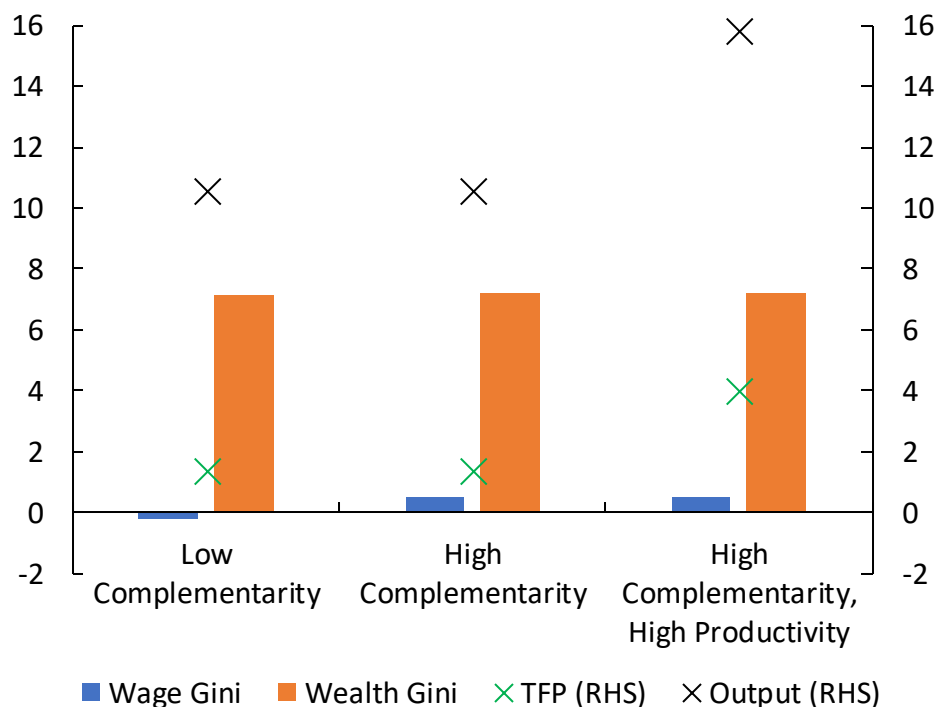
Sources: Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC); UK Labour Force Survey (LFS); and IMF staff calculations.

Note: The figures plot the estimated share of employment by age for each exposure category for college and non-college educated workers, according to the calculations described in Annex 3. Country names use International Organization for Standardization (ISO) country codes.

Under the high-complementarity-high-productivity scenario, the increase in total national income is largest and benefits all workers, although gains are larger for those at the top.

Impact on Aggregates

(Percentage Point on LHS; Percent on RHS)



Sources: IMF Staff calculations.

Note: The figure shows the change in the aggregate wage and wealth Gini between the initial and final distribution in each scenario, as well as the change TFP and output. For more details on the model see SDN Annex 4. TFP = total factor productivity.

- **Scenario 1: Low AI Complementarity**
 - Output increases by nearly 10%
- **Scenario 2: High AI Complementarity**
 - Sectoral shift towards high-complementarity occupations.
 - Income increase is similar to first scenario; wage inequality rises.
- **Scenario 3: High Productivity Impact**
 - Output surges by 16%.
 - Income level rises for all workers

Dimensions of AI Preparedness & Data

- **AI Preparedness Index (AIIPI)** measures readiness across multiple strategic AI adoption areas.
- Builds on cross-country technology diffusion and adoption literature (Keller, 2004; Nicoletti et al., 2020).
- Index includes macro-structural indicators under **four themes**:

Foundational preparedness

1. **Digital infrastructure:** basis for AI tech diffusion and application.
2. **Human capital and labor market policies:** digital skill distribution and policies for labor transitions.

Second-generation preparedness

3. **Innovation and economic integration:** promotes R&D and global trade, attracting investments.
4. **Regulation and ethics:** legal framework's adaptability and governance for enforcement.

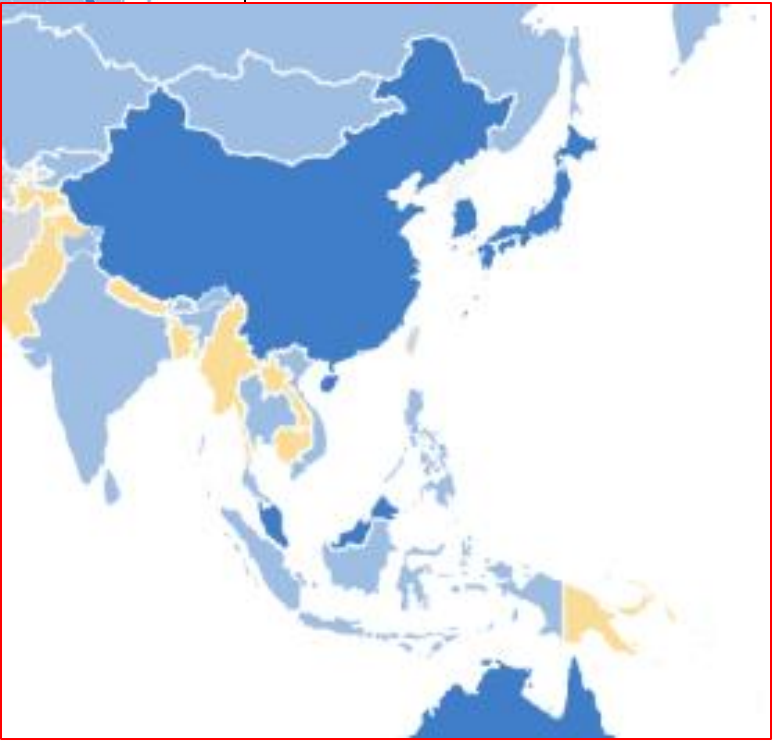
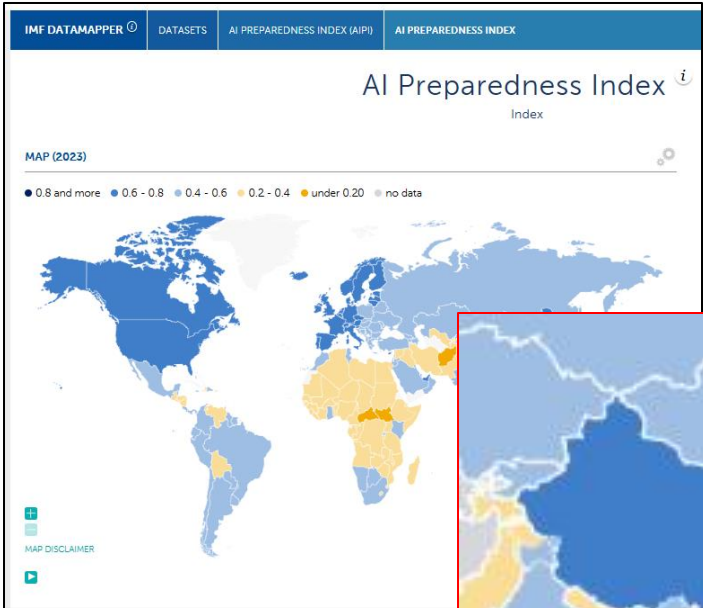
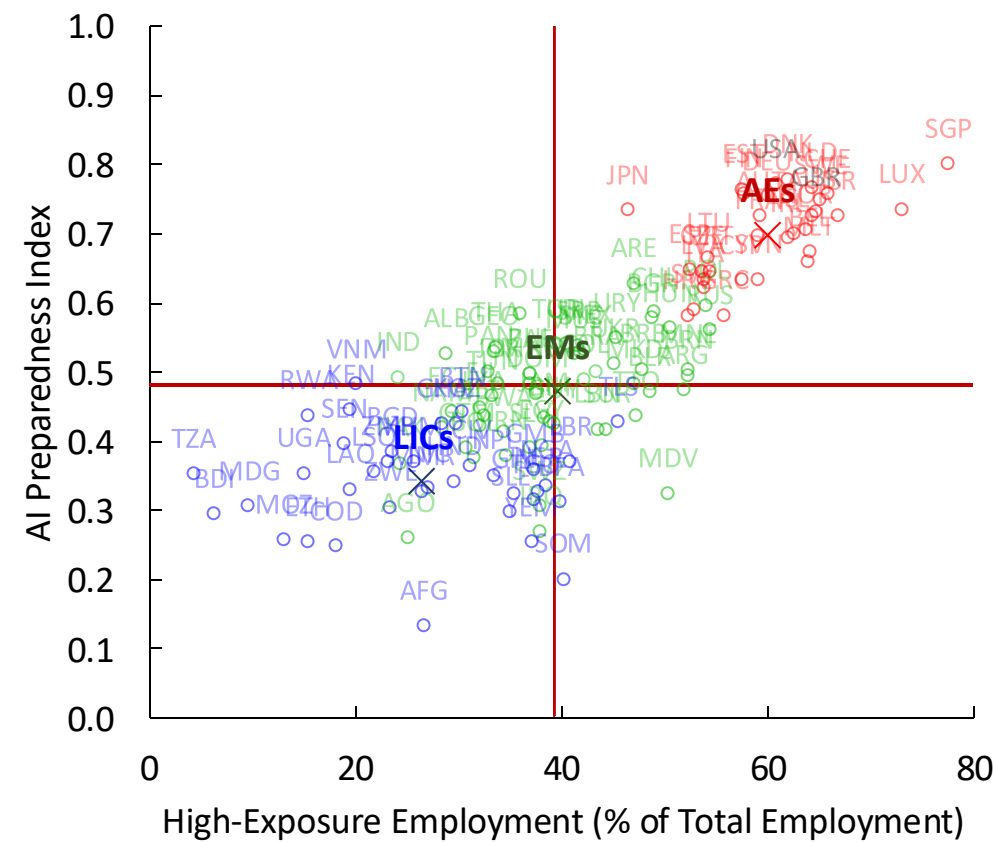
Main Sources of Data for the AIIPI

- International Labor Organization
- International Telecommunication Union
- United Nations
- Universal Postal Union
- World Bank
- World Economic Forum
- Fraser Institute

Dimension	Indicator
1. FOUNDATIONAL AI PREPAREDNESS	
I. Digital Infrastructure	
<i>Accessible, Affordable, and secured Internet Access</i>	- Estimated internet users per 100 inhabitants [UN] - Number of main fixed telephone lines per 100 inhabitants [UN] - Number of mobile subscribers per 100 inhabitants [UN] - Number of fixed broadband subscriptions per 100 inhabitants [UN] - Number of wireless broadband subscriptions per 100 inhabitants [UN] - Cost of internet access (percent of monthly GNI per capita) [ITU] - Secure Internet servers per 1 million people [WB]
<i>Mature e-commerce infrastructure</i>	- Private Sector's E-commerce Business Environment - o Postal Reliability Index [UPU] - o Use of mobile phone for online transactions (% of population ages 15+) [WB] - Public Sector's Online services Infrastructure [UN]
II. Human Capital and Labor Market Policies	
<i>Education and digital skills</i>	- Human Capital Index (i.e., mean years of schooling; expected years of schooling; gross enrolment ratio; adult literacy) [UN] - Public Education Expenditure (10-year average; %GDP) [WB] - Skillset of graduates (proxy for equality of education) [WEF] - Digital skills among active population (e.g., computer skills; basic coding, etc.) [UN] - Number of STEM graduates (10-year average; % of total graduates) [WB] - Number of female STEM graduates (10-year average; % of STEM graduates) [WB]
<i>Labor Market Flexibility & Policies</i>	- Flexibility of wage determination (centralized vs individual firm level) [WEF] - Social protection (% of population covered by social protection schemes) [ILO] - Internal labor market mobility [WEF] - Active labor market policies (e.g., skills matching, retraining) [WEF] - Pay and productivity (i.e., extent to which wages are market determined) [WEF]
2. SECOND-GENERATION AI PREPAREDNESS	
III. Innovation & Economic Integration	
<i>Innovation</i>	- R&D spending per unit of GDP [WB] - Frontier Technology Readiness (i.e., AI related R&D activity: number of scientific publications; number of patents on frontier technologies) [UN] - Domestic credit to private sector (%GDP) [WB]
<i>Economic Integration</i>	- Mean Tariff Rate [FI] - Non-tariff barriers [FI] - Free movement of capital and people (average of three indicators: financial openness; capital controls, and freedom of foreigners to visit) [FI]
IV. Regulation and Ethics	
<i>Strong legal frameworks and enforcement mechanisms</i>	- Legal framework's adaptability to digital business models [WEF] - Overall governance (proxy for enforcement/accountability) [WB]

Higher-income economies, including AEs and some EMs, are generally better prepared than LICs to adopt AI

AI Preparedness Index and Employment Share in High-Exposure Occupations



Sources: International Labour Organization (ILO); and IMF staff calculations.
Note: The plot includes 125 countries: 32 AEs, 56 EMs, and 37 LICs. The red reference lines are derived from the median values of the AI preparedness index and high-exposure employment. Circles represent the average values for each respective country group. Crosses denote the average values for each corresponding country group AEs = advanced economies; EMs = emerging markets; LICs = low-income countries. Country labels use International Organization for Standardization (ISO) country codes.

We estimate the domestic and global impact of AI-induced productivity gains

- Combine data on **sectoral-level jobs' exposures to AI** with **country-specific measures of AI preparedness** and assumptions on **AI access** to gauge their impact on total factor productivity (TFP).
- TFP measures are used as inputs into the **IMF's Global Integrated Monetary and Fiscal model (GIMF)** – a multi-sector, multi-region global model.
- Evaluate their **global and regional impact** on growth and other macroeconomic outcomes.

Scenarios

1. Baseline Scenario

- ▶ All regions have **full access** to AI technologies and data ($AI\ Access_i = 1$)
- ▶ **Heterogeneous productivity gains** reflect existing differences in AI exposure and preparedness

2. Limited AI Access Scenario

- ▶ **EMs (excluding China) and LICs** face restricted access due to geopolitical and export barriers
- ▶ TFP growth in these regions is **reduced by 50%** ($AI\ Access_i = 0.5$ for EMA, EML, LIC)

3. Enhanced AI Preparedness Scenario

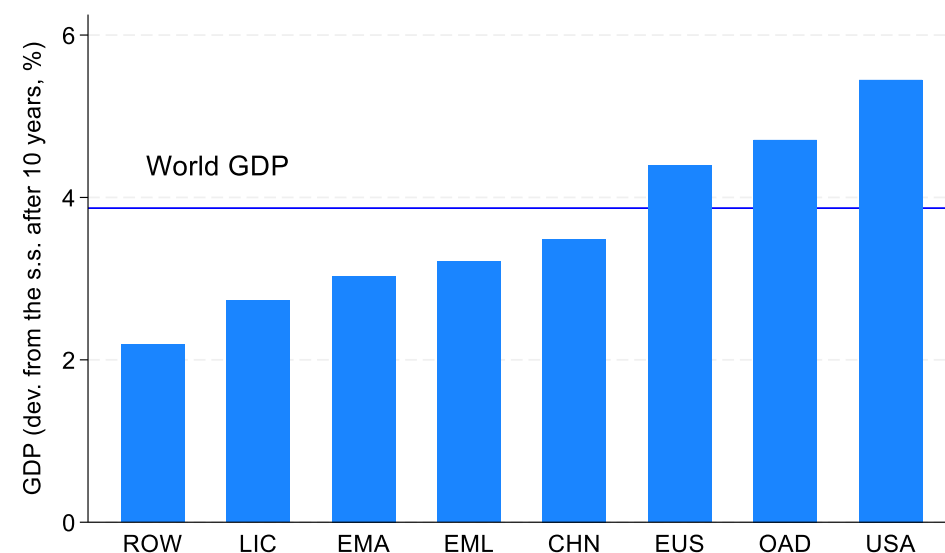
- ▶ **Policy reforms and international support** raise preparedness in EMs and LICs
- ▶ Regions converge toward **peer-best readiness**, enhancing AI-driven TFP gains

Each scenario is examined under the **two TFP growth assumptions**.

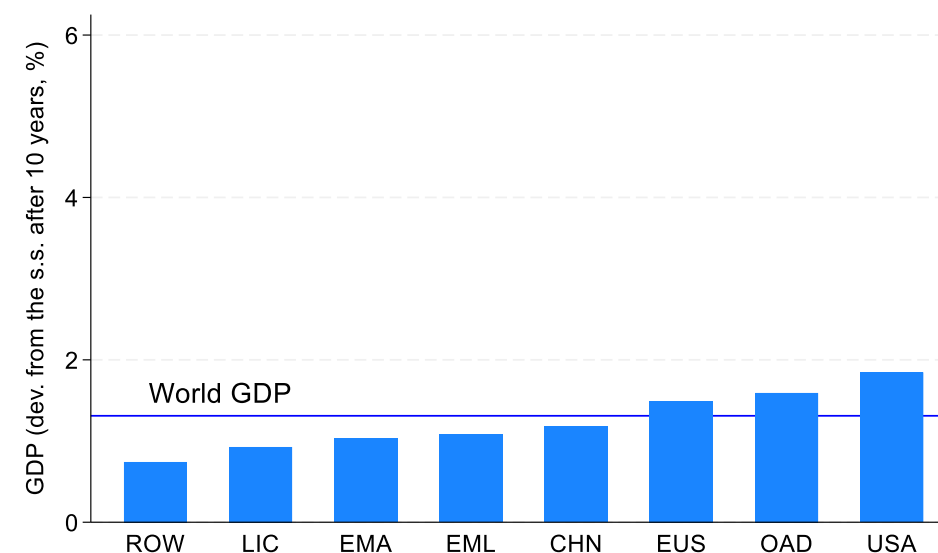
The global impact of the AI shock masks significantly uneven effects

Cross-Country Differences in GDP in the Baseline Scenarios (10-Year Horizon)

High TFP Scenario



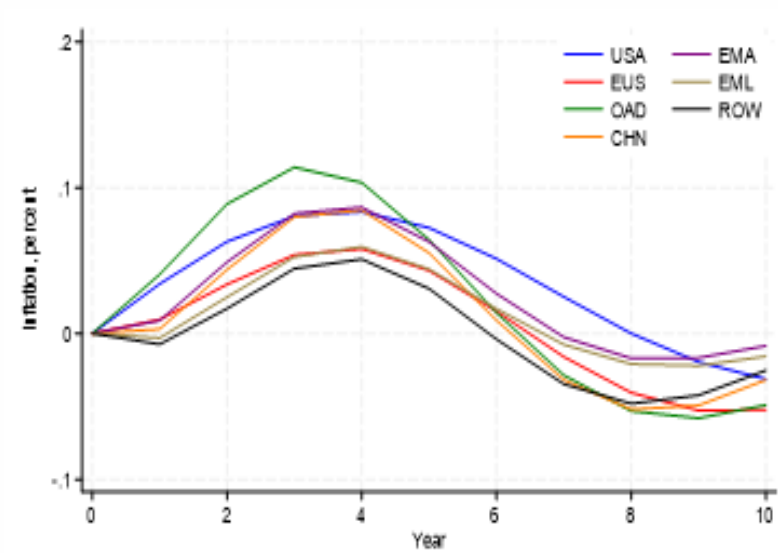
Low TFP Scenario



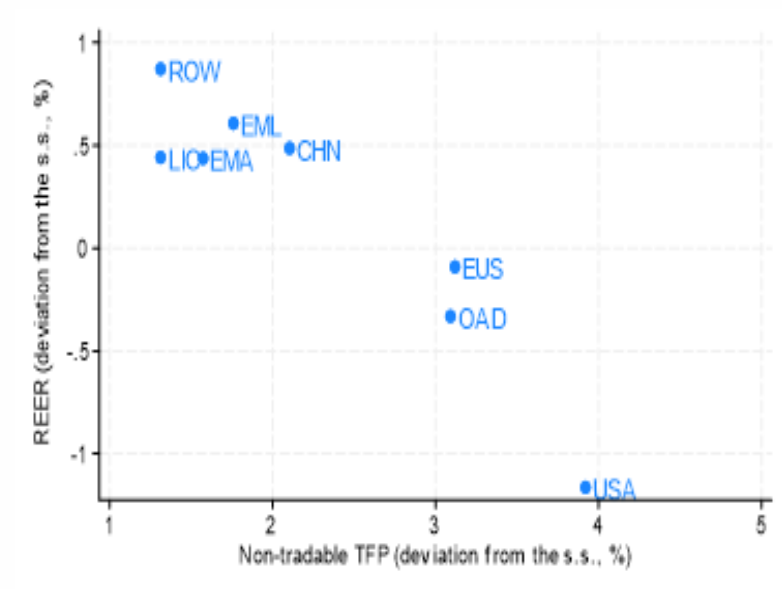
Note: The left panel show the deviations of real GDP from the steady state for the high TFP growth baseline scenario after 10 years. The right panel show the deviations of real GDP from the steady state for the low TFP growth baseline scenario after 10 years. The global averages are shown as horizontal lines. EMA = Emerging Market Economies Asia, Central, Asia, Russia, etc. EML = Emerging Market Economies Latin America, Middle East, Africa, etc. EUS = EU and Switzerland, LIC = Low-income countries, OAD = Other advanced economies and ROW = Rest of the World. The detailed composition of the countries in each group is presented in Annex II.

Some inflationary effects could be observed in the medium term, and AEs' currencies may depreciate

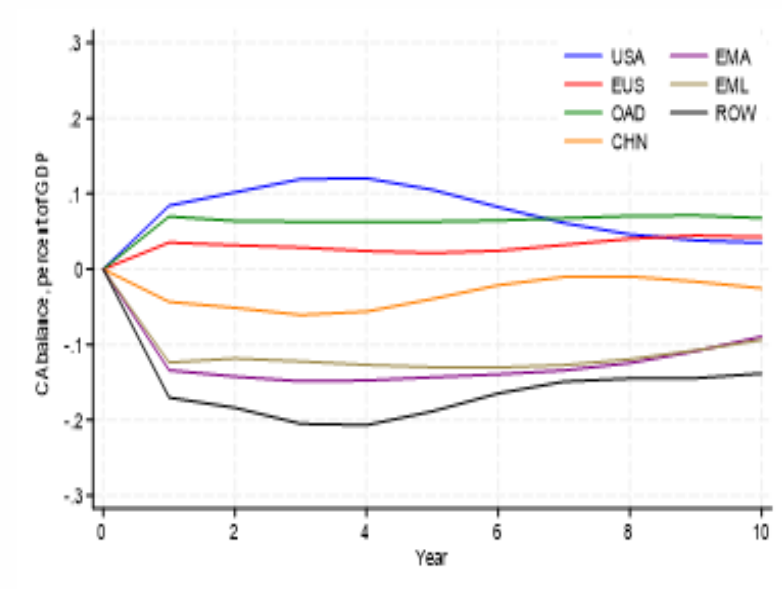
Inflation in the High TFP Growth baseline scenarios



Changes in Real Effective Exchange Rates and Non-tradable Sector TFP (10-year horizon)



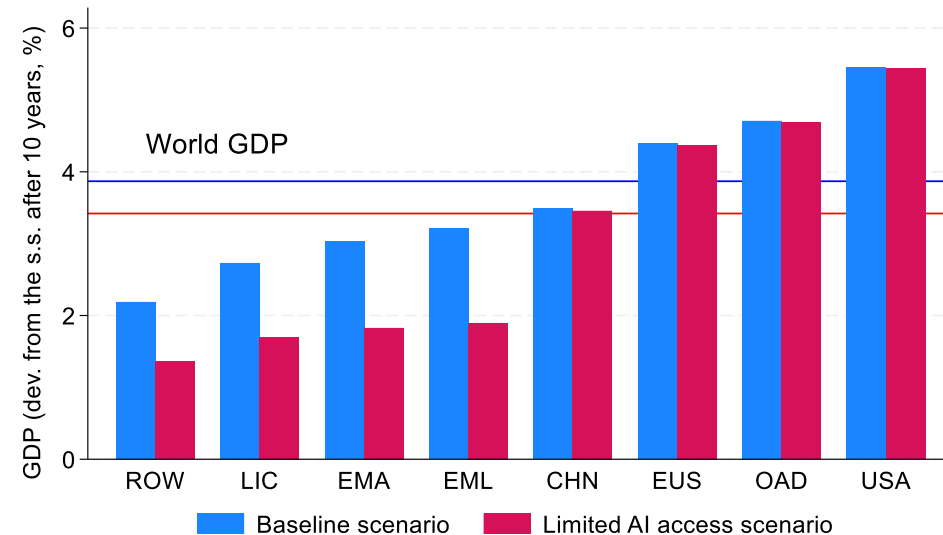
Current Accounts in the High TFP Growth Baseline Scenario



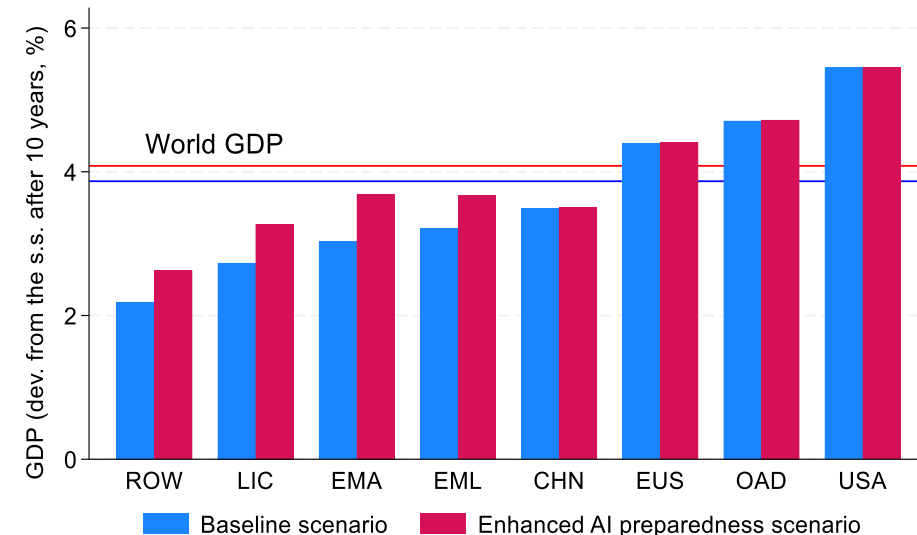
Note: The values are shown as deviations of high TFP growth scenario outcomes from the steady state. See Annex I for the list of ISIC sectors included in the Non-tradable sector. EMA = Emerging Market Economies Asia, Central, Asia, Russia, etc. EML = Emerging Market Economies Latin America, Middle East, Africa, etc. EUS = EU and Switzerland, LIC = Low-income countries, OAD = Other advanced economies and ROW = Rest of the World. The detailed composition of the countries in each group is presented in Annex II.

AI access and preparedness make a significant difference for EMDEs; unlikely to close the gap

Regional Differences in the Limited AI Access Scenario



Regional Differences in the Enhanced AI Preparedness Scenario



Note: The left panel show the deviations of real GDP from the steady state for the high TFP growth baseline scenario after 10 years. The right panel show the deviations of real GDP from the steady state for the low TFP growth baseline scenario after 10 years. The global averages are shown as horizontal lines. EMA = Emerging Market Economies Asia, Central, Asia, Russia, etc. EML = Emerging Market Economies Latin America, Middle East, Africa, etc. EUS = EU and Switzerland, LIC = Low-income countries, OAD = Other advanced economies and ROW = Rest of the World. The detailed composition of the countries in each group is presented in Annex II.

Conclusions

Conclusions

Almost 40% percent of global employment is exposed to AI

- 60% of AE jobs are exposed to AI, 40% in EMs, and 26% in LICs, mostly cognitive roles.
- AI may negatively affect half of these jobs; the other half could gain productivity.
- AEs are generally at greater risk but also better poised to exploit AI benefits than EMDEs.

AI may lead to higher income and wealth inequality

- AI complementarity is highly correlated with income.
- AI may increase capital returns.

Mind the gap: AEs likely to benefit disproportionately

- The estimated growth impact in AEs more than double than in LICs.
- Policies may not fully close the gap; some could aggravate disparities.

To harness AI's potential fully, priorities depend on countries' development levels

- AEs and some EMs are ahead in AI readiness compared to LICs.
- AEs and some EMs should focus on AI regulation and invest in AI innovation and integration.
- EMDEs need digital infrastructure and training.

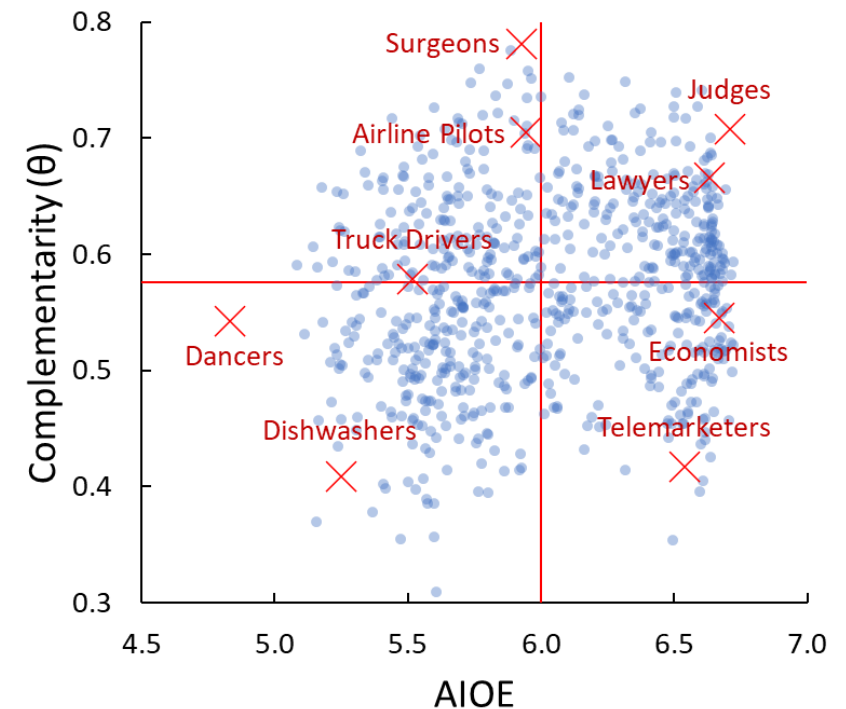
Thank you!

Annex

Measuring exposure to and complementarity with AI

- **Exposure to AI:** Degree of overlap between AI applications and human abilities in occupations (Felten et al., 2021;2023).
- **Potential complementarity Index:** Leverage two parts of the O*NET capturing “work context” and “skills.” Group into 6 categories:
 - **a. Communication:** Face-to-Face, and public Speaking
 - **b. Responsibility:** Responsibility for outcomes and others’ health
 - **c. Physical Conditions:** Outdoors exposed, and physical proximity
 - **d. Criticality:** Consequence of error, freedom and frequency of Decisions
 - **e. Routine:** Degree of automation, and unstructured vs structured Work
 - **f. Skills:** Job zone (level of education, training and skills needed)
- **Examples:**
 - Judges: High AI exposure yet shielded by societal norms and laws—AI may complement their work, enhancing productivity.
 - Clerical Workers: High AI exposure with low shielding—higher displacement risk.

Conceptual Diagram of AI Exposure (AIOE) and Complementarity (θ)



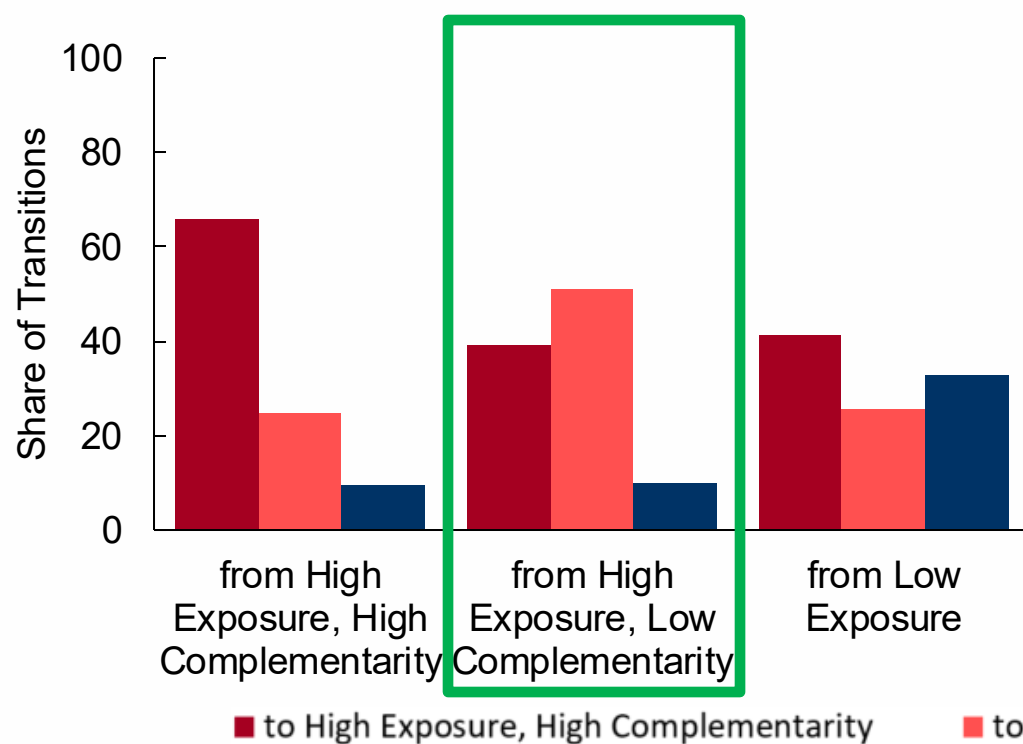
Sources: Felten, Raj, and Seamans (2021); Pizzinelli and others (2023); and IMF staff calculations.

Note: Red reference lines denote the median of AIOE and complementarity.

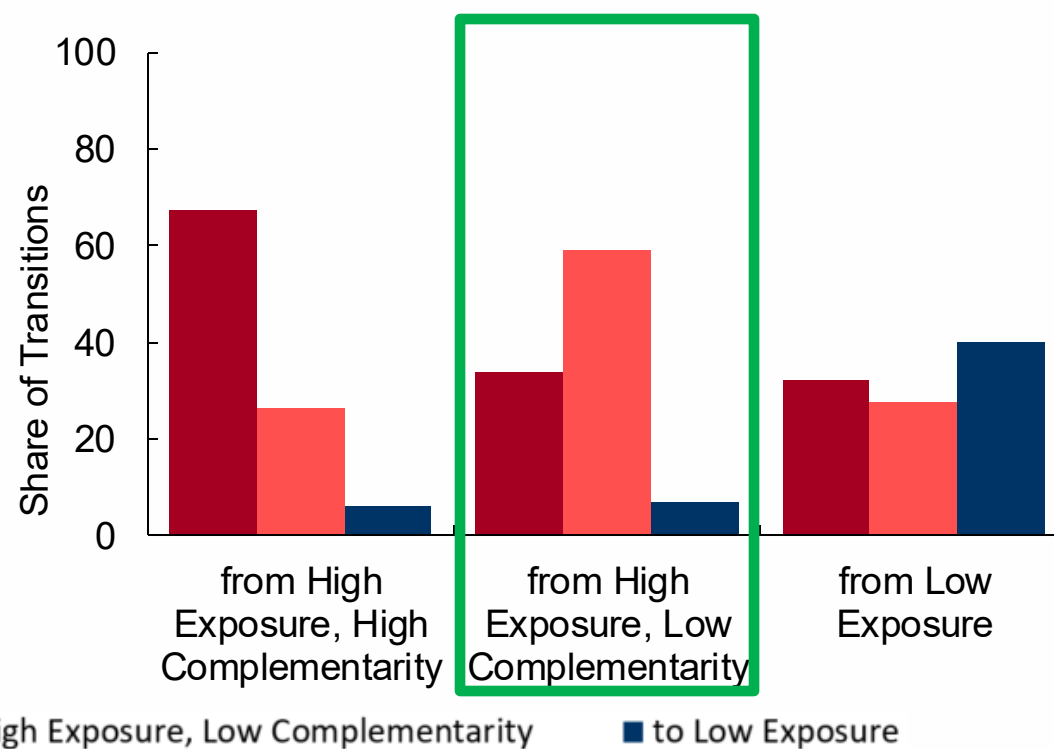
Workers with college education have historically shown a greater ability to transition into what are now jobs with high AI-complementarity potential

Occupational Transitions for College-Educated Workers

1. BRA
(Percent)



2. GBR
(Percent)



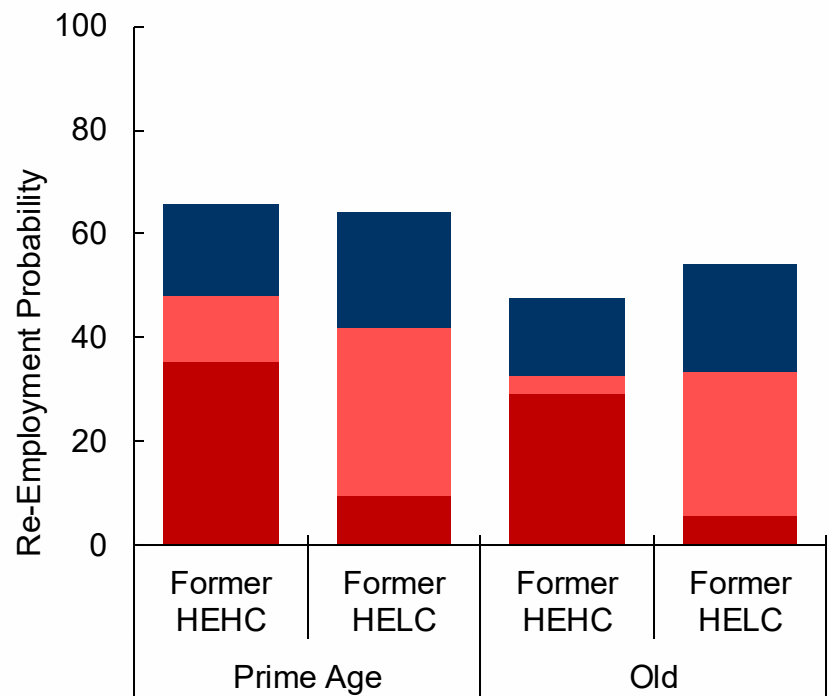
Sources: Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC); UK Labour Force Survey (LFS); and IMF staff calculations.

Note: "From" indicates the exposure category of the occupation the individual had in the preceding quarter, while "to" indicates the exposure category of the occupation the worker transitioned to. The share of transitions represents the average share of transitions in the "from" category for college-educated workers that go to the "to" category. Country names use International Organization for Standardization (ISO) country codes.

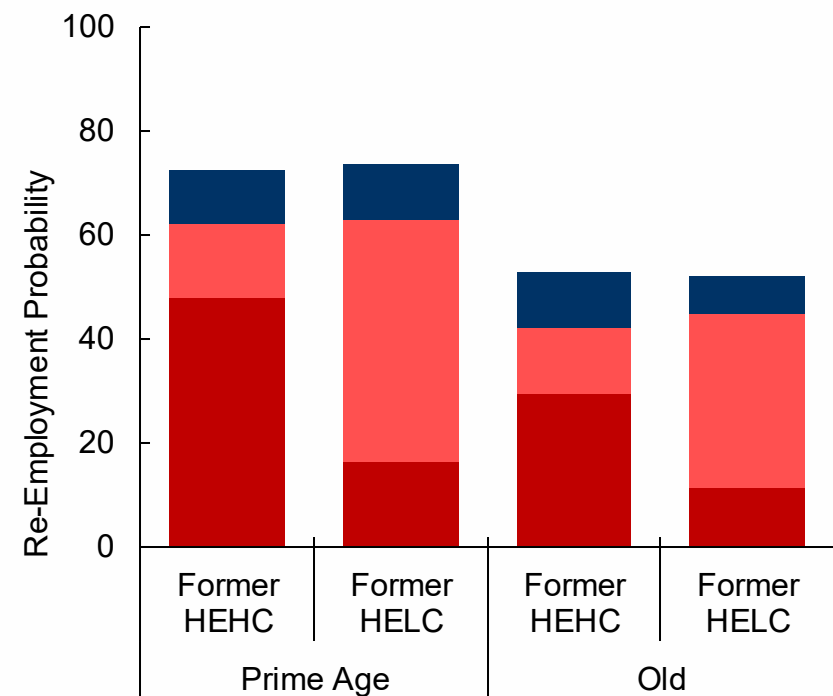
Older workers may be less adaptable, as reflected in their lower likelihood to be re-employed after termination

1-Year Re-Employment Probability of Separated Workers

1. BRA



2. GBR



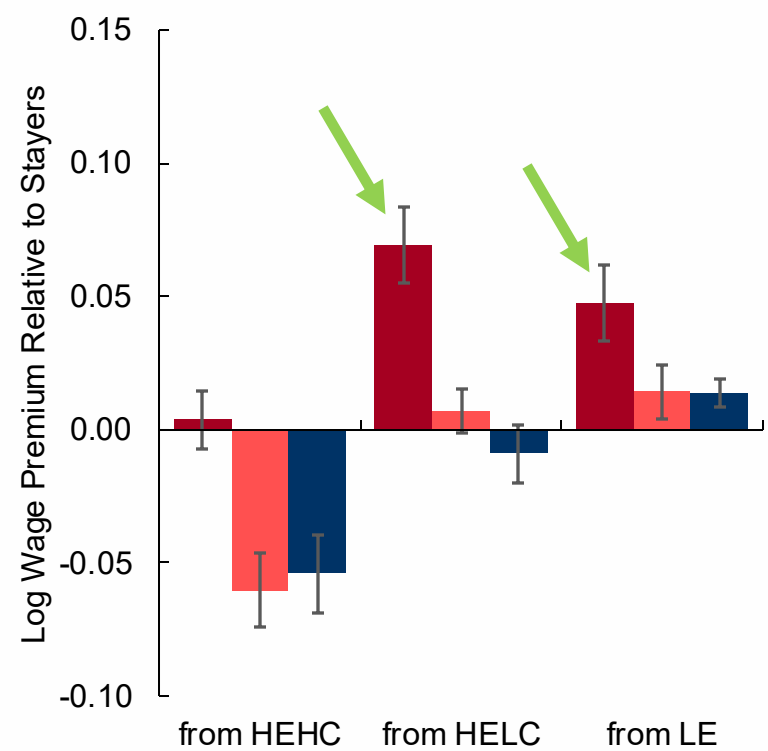
■ to High Exposure, High Complementarity (HEHC) ■ to High Exposure, Low Complementarity (HEL) ■ to Low Exposure (LE)

Sources: Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC); UK Labour Force Survey (LFS); and IMF staff calculations.
Note: The bars report the re-employment probability of workers who have recently (within the last quarter) transitioned from employment to unemployment, which is defined as the share of these workers who are again employed one year later. "From" indicates the exposure category of the occupation the individual had before being unemployed, while "to" indicates the exposure category of the occupation the worker transitioned to. "Prime Age" refers to workers over 35 and under 55, while "old" refers to workers 55 and older. Country names use International Organization for Standardization (ISO) country codes.
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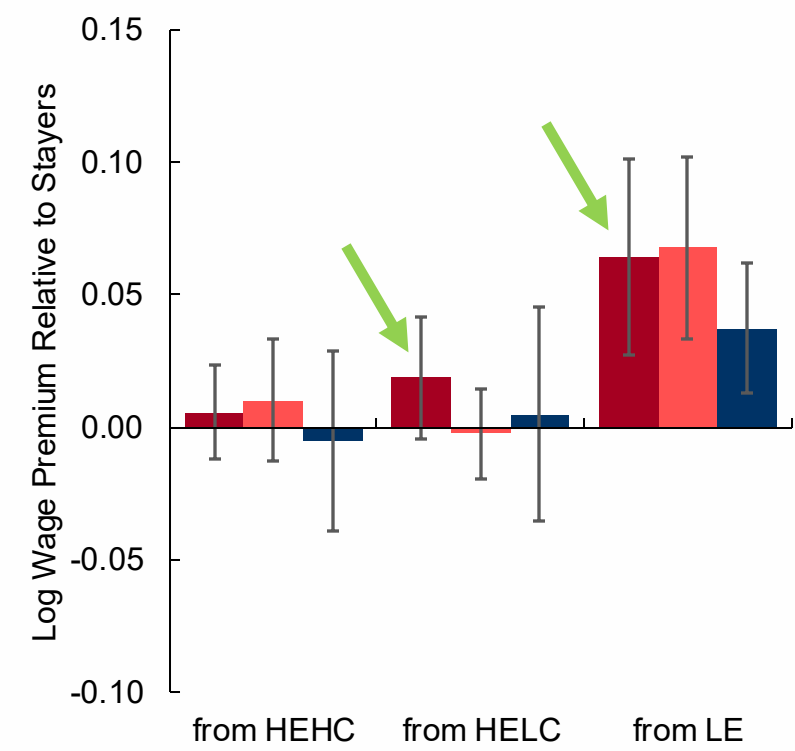
Occupation switches also affect workers' incomes

Estimated Wage Premia from Occupation Changes

1. BRA



2. GBR



■ to High Exposure, High Complementarity (HEHC) ■ to High Exposure, Low Complementarity (HELC) ■ to Low Exposure (LE)

Sources: Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC); UK Labour Force Survey (LFS); and IMF staff calculations.
Note: "From" indicates the exposure category of the occupation the individual had in the preceding year, while "to" indicates the exposure category of the occupation the worker transitioned to. The premia are "relative to stayers", that is, they represent the increase or decrease in wages in relation to workers in the "from" category that did not switch occupations over a year. Wage premia are calculated according to the regression specification in Annex 2. 95% confidence intervals for the point estimates are shown in bars. Country names use International Organization for Standardization (ISO) country codes.
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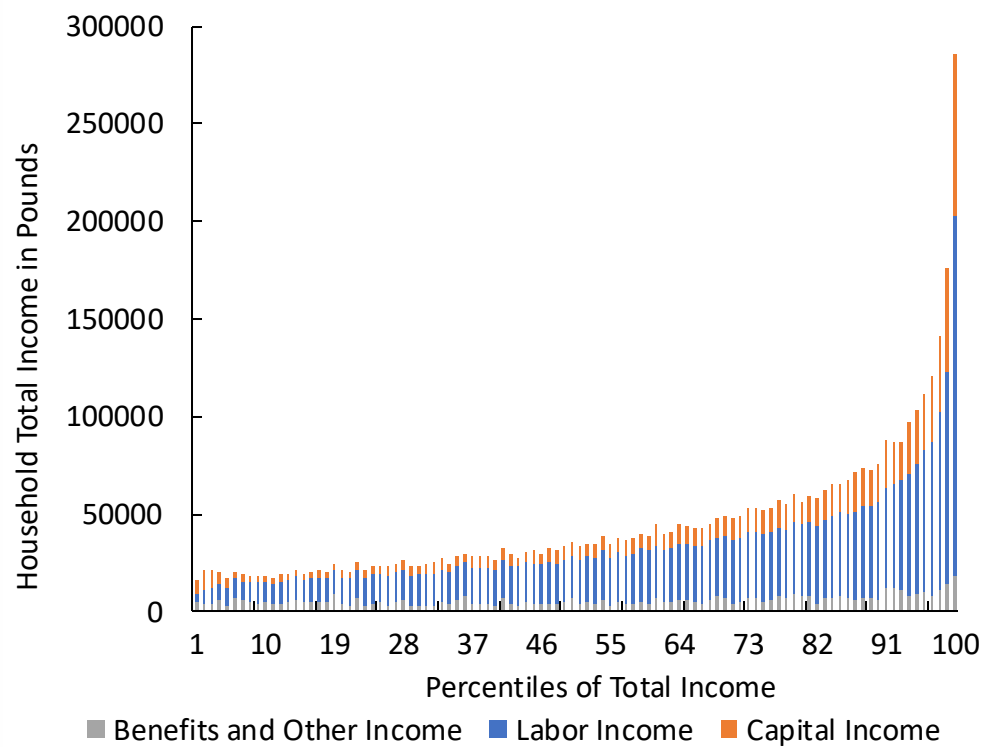
Model-based analysis of AI's economic impact

- **Task-based model** by Rockall, Pizzinelli and Tavares (2023) assesses effects on income distribution and wider economic impacts stemming from AI adoption. The Model builds on the work of Moll, Rachel and Restrepo (2023) and Drozd, Taschereau-Dumoucheland, and Tavares (2023)
- Model incorporates differences in labor productivity, asset holdings, AI exposure, and complementarity.
- **Four critical channels of impact of AI** are identified:
 1. **Labor displacement:** Shift of tasks from human labor to AI capital, reducing labor income.
 2. **Complementarity:** Value added shifts to AI-complementary occupations, increasing labor demand for these occupations and reducing it for others.
 3. **Capital income:** AI adoption leads to increases in the return of capital, raising capital income further.
 4. **Productivity gains:** Overall economic boost potentially offsets labor income losses.
- Calibration to the UK Economy.

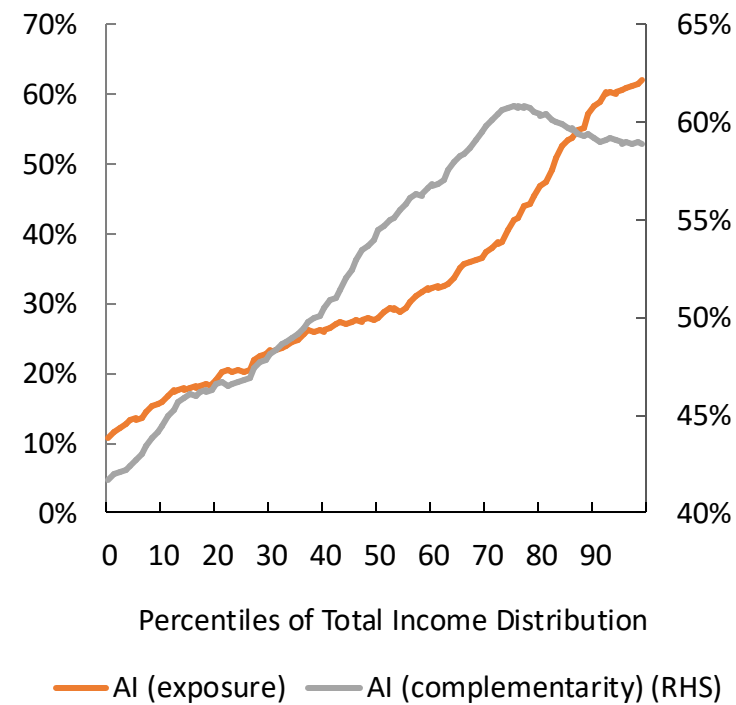
While middle and low-income workers' total income depends mostly on wage income, high-income workers have a large share of capital income.

Exposure to AI and to Automation and Income in the UK

1. Exposure of Income to AI
(Pounds Sterling)



2. Exposure and Complementarity by Income Percentiles
(AI and Complementarity Index)

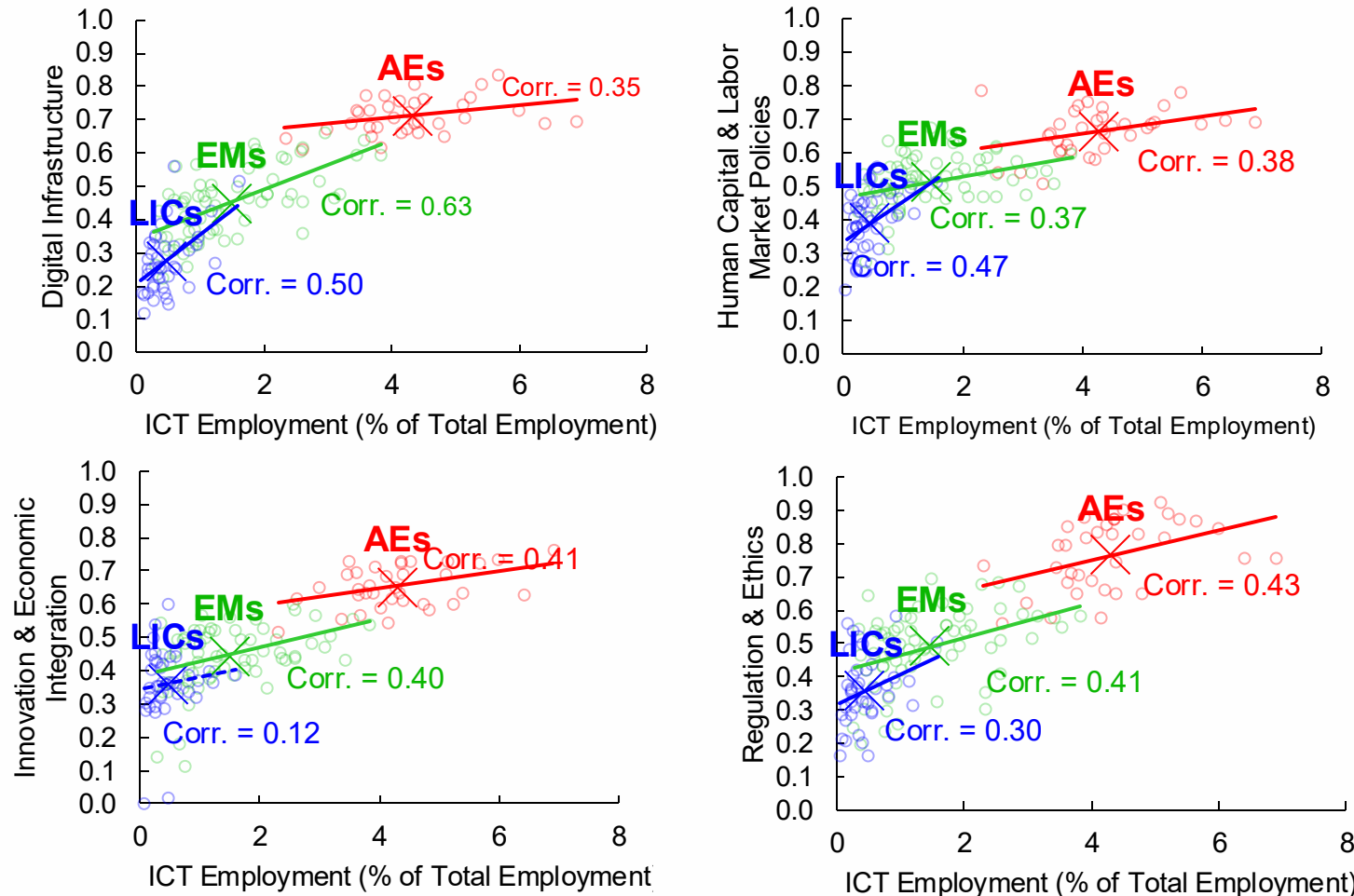


Sources: Wealth and Assets Survey (WAS); and IMF Staff calculations.
Note: Plot 1 reports three categories of workers' income by total income percentiles: (i) wage income, (ii) benefits, pensions, and other income, and (iii) capital income (rents and estimated investment income). In plot 2, AI exposure is measured as the share of total hours worked in a job in the top 30% of AI Occupational Exposure (AIOE) scores, from Felten, Raj, and Seamans (2021), weighted by hours worked. This threshold is chosen to make the analysis comparable to historical episodes of automation. AI complementarity is measured by considering the work contexts and skills, as discussed in Box 1 and in detail in Pizzinelli and others (2023). In the panel, we plot AI exposure and complementarity by total income percentiles.

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Reform prioritization should align with AI preparedness gaps, which vary across the development spectrum

ICT Employment Share and Individual Components of the AI Preparedness Index



Policy prioritization should distinguish between:

- **Foundational AI preparedness** (digital infrastructure and human capital that enable workers and firms for AI adoption) is crucial for LICs and many EMs.
- **Second-generation preparedness** (innovation and legal frameworks) is crucial for AEs (and some EMs) with already strong foundational preparedness and digital skills.

Sources: International Labour Organization (ILO); and IMF staff calculations.

Note: ICT employment refers to people working in the information and communication sector based on ISIC-Rev 4 classification. 142 countries are included: 35 AEs, 67 EMs, and 40 LICs. Circles represent the average values for each respective country group. Crosses denote the average values for each corresponding country group. Simple correlation ("Corr.") is also added for each country group. AEs = advanced economies, EMs = emerging markets, LICs = low income countries, ISIC = International Standard Industrial Classification.

Within all regions, AI exposure follows a sectoral pattern: AI-intensive > non-tradables > tradables

Country groups	Abbreviation	Sector	AI exposure	AI preparedness
China	CHN	Tradable	0.30	0.64
		Non-tradable	0.43	
		AI-intensive	0.58	
Emerging Market Economies Asia, Central Asia, Russia, etc.	EMA	Tradable	0.18	0.50
		Non-tradable	0.41	
		AI-intensive	0.72	
Emerging Market Economies Latin America, Middle East, Africa, etc.	EML	Tradable	0.24	0.50
		Non-tradable	0.46	
		AI-intensive	0.72	
EU and Switzerland	EUS	Tradable	0.40	0.68
		Non-tradable	0.60	
		AI-intensive	0.82	
Low-income countries	LIC	Tradable	0.17	0.35
		Non-tradable	0.47	
		AI-intensive	0.72	
Other advanced economies	OAD	Tradable	0.36	0.73
		Non-tradable	0.56	
		AI-intensive	0.77	
United States	USA	Tradable	0.43	0.77
		Non-tradable	0.67	
		AI-intensive	0.85	

The Methodology: TFP Shocks

- **Region-level TFP** is calculated as proportional to their AI preparedness and AI exposure relative to the benchmark (the UK), interacted with AI access.
- **Sector-level TFP** is then calculated based on the size of region-level TFP and sector-level AI exposure (relative to region-level AI exposure).

Regional TFP: $TFP\ shock_{i,t} = \frac{AI\ preparedness_i \times AI\ exposure_i}{AI\ preparedness_b \times AI\ exposure_b} \times AI\ Access_{i,t} \times TFP\ shock_{b,t}$

Sectoral TFP: $TFP\ shock_{i,j,t} = \frac{AI\ exposure_{i,j}}{AI\ exposure_i} \times TFP\ shock_{i,t}$

i : region; j : sector; t : time

Benchmark TFP Shocks

- **Literature.** Great uncertainty about AI's TFP impact, e.g., for the U.S.:
 - ▶ **LOW: Acemoglu (2025)** estimate TFP gains of c. 0.7% over the next 10 years.
 - ◆ This estimate hinges on 4.6% of tasks being replaced profitably by AI, and a 27% increase in productivity due to labor cost savings.
 - ▶ **HIGH: Aghion and Bunel (2024)** median estimate of 6.8% over 10 years.
 - ◆ Here (i) the share of tasks exposed to AI in c. 60%; (ii) the share of exposed tasks for which it will be profitable to use AI c. 50%; (iii) labor saving costs induced by AI of 40%.
- **This paper.** Uses two benchmark TFP scenarios based on UK simulations (Cazzaniga et al. 2024), scaled by country-specific AI exposure & preparedness.
- Resulting **US-wide** TFP gains over 10 years:
 - ▶ **Low TFP benchmark:** 1.3% ($\approx 2\times$ Acemoglu)
 - ▶ **High TFP benchmark:** 3.4% (below Aghion & Bunel)